

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Describe utilization of grog, blast furnace slag, fly ash in making refractories.
- Q.24 Describe the refractories-used in different parts of Glass Tank FURNACE with the help of neat sketch.
- Q.25 Describe Castable, classification of castable and advantages of monolithic / castable refractories.

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Roll No.

5th Sem / Ceramic

Subject : Refractory Applications

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Segar Cones is also called as _____.
- a) Pyrometric cones b) Pyrometric Qones
c) Pyrometric Cones d) Pyrometric Tones
- Q.2 Insulating refractory has _____.
- a) Low permeability
b) High specific gravity
c) Low porosity
d) High porosity
- Q.3 The term “Cermet” is a combination of ceramic and _____
- a) Wood b) metal
c) Plastic d) clay

Q.4 Which of the following is a slow rise of plastic deformation under the action of shear stresses when it is below the yield strength of the material?

- a) Brittle fracture b) Ductile fracture
- c) Creep d) Fatigue

Q.5 Tank furnace is used to make _____

- a) Steel b) iron
- c) glass d) tile

Q.6 What is hearth in a furnace?

- a) The electrodes are often called hearth
- b) The bowl-shaped bottom of the furnace
- c) The walls of the furnace
- d) The dome shaped roof of the furnace

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Monolithic means multiplayer. (True/False)

Q.8 SiC refractories are used in making of crucibles. (True/False)

Q.9 Porosity of insulating refractory brick should be high. (True/False)

Q.10 Cermet's are used in making of cutting wheels. (True/False)

Q.11 PLC means _____.

Q.12 Alumina-Silica phase diagram is two component system. (True/False)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain Crucible.

Q.14 Explain Burner blocks.

Q.15 Discuss Cement Rotary kiln.

Q.16 Explain refractories used in basic open-hearth furnace.

Q.17 Discuss lining pattern and type of refractory used in EAF steel making.

Q.18 Explain refractories used in Reheating furnace.

Q.19 Explain refractories in tunnel kiln.

Q.20 Discuss types of refractory in lime and dolomite kilns.

Q.21 Explain Spalling resistance.

Q.22 Explain HOT MOR test.